



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CC8X94 - NEXT GENERATION WIRELESS NETWORK

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

	Unit – I	Marks	BT*	CO*	PO*
1.	a) Briefly describe the 5G standardization activities for ITU, 3GPP and IEEE.	10	L*2	1	1
	b) Explain the challenging that has to be fulfilled by 5G systems for the collection of diverse use cases in present scenario.	10	L2	1	1
2.	a) Illustrates the main features of the 3GPP standards in the market, highlighting the trend toward widespread use of spectrum, higher bandwidths, higher spectral efficiency and lower latency.	10	L2	1	1
	b) Explain the Software Defined Networking (SDN) for 5 G systems with diagram and highlights its benefits.	10	L2	2	1
3.	a) Describe how the placement of functionality may be determined by the type of deployment.	10	L2	2	1
	b) Illustrate the integration of LTE and new air interface to ful-fill 5G requirements and draw the diagram.	10	L2	2	1
	Unit – II				
4.	a) Explain the uplink transmission using coded random access with 5 users u_i and a set of 2 pilot sequences s_1 and s_2 .	10	L2	3	1
	b) Explain the beam steering control of antenna blocks with hybrid beam-forming.	10	L2	3	1
5.	a) Explain the Basic forms of massive MIMO implementation with block diagrams.	10	L2	3	1
	b) Discuss the iteration algorithm for spectrum allocation in multi-operator D2D communication.	10	L2	4	1
6.	a) Using suitable plots discuss the performance of D2D communications in the proximity communications scenario.	10	L2	4	1
	b) Explain the performance analysis for 5G D2D RRM concept with suitable examples.	10	L2	4	1
	Unit – III				
7.	a) Discuss the design and implementation for Samsung 60 GHz Wi-Fi.	10	L2	5	1
	b) Discuss the challenge of directional communication in practical indoor environments and principle of antenna beamforming.	10	L2	5	1
8.	a) Elaborate the Wireless Paradigms and Frequencies Above 100GHz.	10	L2	5	1
	b) Explain the Multiuser operation for WiFi6.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
